Library Management System — UML Class Diagram Report

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1. Introduction

A UML class diagram shows the classes of a system, their attributes and methods, and the relationships between them. This report presents the class diagram for the Library Management System. It contains four classes: **Book**, **Member**, **Librarian**, and **Loan**.

2. UML Class Diagram

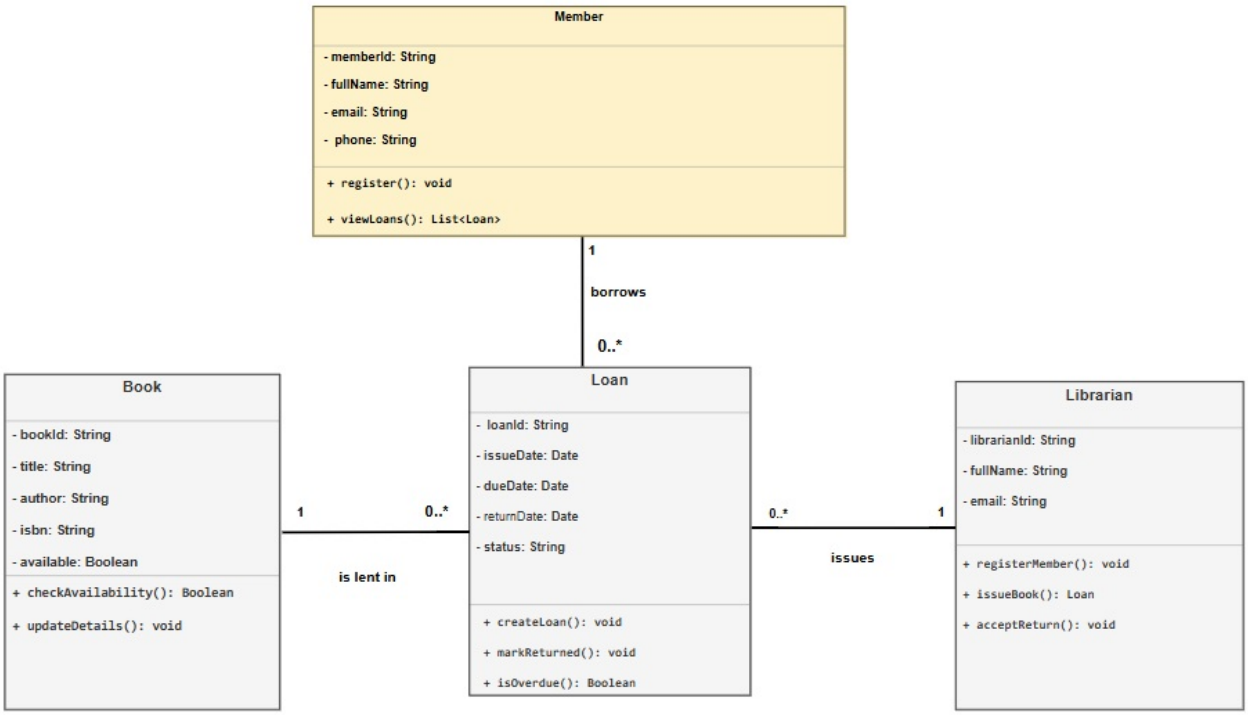


Figure 1: UML class diagram of the Library Management System.

3. Classes, Attributes, and Methods

3.1 Book

Attributes	Methods
`- bookId: String`	`+ checkAvailability(): Boolean`
`- title: String`	`+ updateDetails(): void`
`- author: String`	

`- isbn: String`	
`- available: Boolean`	

Stores information about a library book. The `available` attribute shows whether the book can currently be borrowed.

3.2 Member

Attributes	Methods
`- memberId: String`	`+ register(): void`
`- fullName: String`	`+ viewLoans(): List<Loan>`
`- email: String`	
`- phone: String`	

Stores information about a registered library user. A member can register and view their own loans.

3.3 Librarian

Attributes	Methods
`- librarianId: String`	`+ registerMember(): void`
`- fullName: String`	`+ issueBook(): Loan`
`- email: String`	`+ acceptReturn(): void`

Represents the employee who manages library activities, including registering members, issuing books, and accepting returns.

3.4 Loan

Attributes	Methods
`- loanId: String`	`+ createLoan(): void`
`- issueDate: Date`	`+ markReturned(): void`
`- dueDate: Date`	`+ isOverdue(): Boolean`
`- returnDate: Date`	
`- status: String`	

Represents one borrowing transaction. It records the dates of borrowing and returning and the current status of the loan.

4. Relationships and Multiplicities

All three relationships are **associations** with the Loan class at the centre.

Relationship	Label	Multiplicity	Explanation
Member — Loan	borrows	**1** to **0..***	One member can borrow zero or many times. Each loan belongs to exactly one member.
Book — Loan	is lent in	**1** to **0..***	One book can appear in zero or many loans over time. Each loan is for exactly one book.
Librarian —	issues	**1** to **0..***	One librarian can issue zero or many loans. Each loan is

Loan			issued by exactly one librarian.

5. Design Notes

- `register()` in Member lets a member sign up, while `registerMember()` in Librarian lets the librarian register a member on their behalf.
- `issueBook()` in Librarian is the action of lending a book, and it uses `createLoan()` in Loan to create the loan record.

6. UML Notation Used

- `-` means a private attribute; `+` means a public method.
- `String` is text, `Boolean` is true or false, `Date` is a date, and `void` means nothing is returned.
- `1` means exactly one; `0..*` means zero or many.

7. Conclusion

The class diagram shows how members, books, librarians, and loans are related in the Library Management System. Loan connects the other three classes, so every borrowing transaction can be traced to one member, one book, and one librarian.